



**JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR**



(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATE, RTMNU & MSBTE Mumbai
Basic Science & Humanities Department
2025-26 (Semester)
MID SEM EXAMINATION

Course: B. Tech.

Subject: VEDIC MATHEMATICS (Open Elective)

Subject Code: CS/IT/AI/DS/CE/EE/ME/ET30004

Year/Semester: IIIrd / Semester -V

Max Marks: 20

Date: - 08 / 10 /2025

Duration:- 1 Hr

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Course Outcome At the end of course, Student shall be able to:

CO1: Describe concept of:

Demonstrate a comprehensive understanding of the history, principles, and significance of Vedic Mathematics. Recognize and explain the basic principles and concepts of Vedic Mathematics, including the 16 Sutras and their applications.

CO2: Illustrate the concept of :

Perform addition, subtraction, multiplication, and division mentally using Vedic Sutras with speed and accuracy. Apply mental calculation techniques to solve arithmetic problems efficiently.

CO3: Apply the knowledge of :

Apply advanced Vedic Mathematics techniques for squaring, finding square roots, cube roots, and factorization. Solve algebraic equations and manipulate algebraic expressions using Vedic methods.

CO4: Evaluate the problems:

Apply Vedic Mathematics techniques effectively to solve problems encountered in competitive exams such as aptitude tests, entrance exams, and Olympiads. Develop strategies to improve speed and accuracy in solving mathematical problems.

CO5: Analyze the problems and results:

Apply Vedic Mathematics principles to solve real-life problems in various fields such as finance, engineering, and science. Analyze and solve complex problems using Vedic Mathematics techniques, fostering critical thinking and problem-solving skills.

CO6: Create the methods or model:

Explore advanced topics in Vedic Mathematics, such as continued fractions and exponential series. Engage in research activities to explore modern extensions and applications of Vedic Mathematics in diverse areas.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Tell at least one quadratic equation.	1/CO1	1
Q.2	Tell the value of x for $x^2 - 5x + 6 = 0$.	1/CO1	1
Q.3	Tell the name of at least one Vedic mathematics sutra.	1/CO2	1
Q.4	Outline the at least one Vedic Mathematics sutra in Merger (Paravartya).	2/CO2	1

SECTION B

1. All Questions are compulsory.

2. Solve answer in brief (two, three Sentence)

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| Q.5 | Identify the value of x for $x + \frac{1}{x} = \frac{13}{6}$ by using Vedic Mathematics. | 3/CO3 | 2 |
| Q.6 | Explain the Anurupyena sutra with suitable example. | 2/CO1 | 2 |
| Q.7 | Outline the value of x for $\frac{7}{7x+1} + \frac{6}{4x+1} = \frac{15}{6x+1}$ by using Vedic Mathematics. | 2/CO1 | 2 |
| Q.8 | Explain Paravartya Yojayet sutra. | 2/CO2 | 2 |

SECTION C

1. Solve any two questions.

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|------|---|-------|---|
| Q.9 | Examine the value of x, y for $\begin{matrix} 2x + 3y = 8 \\ 4x + 5y = 14 \end{matrix}$ by using Vedic mathematics. | 4/CO3 | 4 |
| Q.10 | Analyze the value of x for $ax^2 + bx + c = 0$ in Vedic Mathematics. | 4/CO3 | 4 |
| Q.11 | Examine the Complex Mergers with example. | 4/CO4 | 4 |